GEORGE CHRIS CARTER

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SENIOR SOFTWARE ENGINEER | FULL-STACK RUBY ON RAILS

Senior full-stack engineer with nearly five years building and running research, HR and innovation web applications across Mass General Brigham. Grew from junior to lead to senior on a four-person engineering team that runs close to 40 production Rails apps, owning ten of them end to end and working largely independently: took over four legacy systems and carried them from Rails 5 to 8, rewrote two (one from Ember), built three new apps and the team's Okta single-sign-on engine from scratch, and led the move from on-premises hosting to AWS.

SKILLS

Languages & Frameworks: Ruby 4 / Rails 8.1 (apps on 5.2 through 8.1), JavaScript, Stimulus, Hotwire/Turbo, jQuery, HAML, Sass, Bootstrap, esbuild, importmap

Data & Background Processing: MySQL, ActiveRecord, multi-database apps, Redis, Solid Queue, Delayed Job, cron-driven imports and reminders

Testing & Code Quality: RSpec, FactoryBot, Capybara/Selenium, SimpleCov, RuboCop, Brakeman, Cycode, CodeClimate/qlty, HoneyBadger

Delivery & Infrastructure: AWS, Docker, GitHub Actions CI, Capistrano, Puma, Linux VMs, shared Rails engines

Auth & Integrations: LDAP, Okta OIDC (PKCE, JWKS), Microsoft Graph, IdM/HR syncs (PeopleSoft, Workday), REST APIs with token auth, PDF/Excel generation

EXPERIENCE

Mass General Brigham – MGB Digital Resource Operations, Research Informatics Core **Jan 2026 – Present**
Massachusetts General Hospital – Center for Systems Biology, Research Informatics Core **Dec 2021 – Jan 2026**

Systems Engineer II – Senior Developer, Mass General Brigham **Jan 2026 – Present**

- Designed and built a protocol submission and multi-stage review platform solo, from an empty repository to production: versioned form templates, auto-assigned reviews, batched comment release with digest emails, institution-based routing; ~400 protocols from 200+ study teams across 35 institutions in its first ten months.
- Integrated that platform with two downstream clinical-research systems through a push-model REST API (token auth, retrying Faraday client, async sync job) and wrote the receiving-side processors; every approved protocol has synced through it since launch.
- Greenfield build of a facility inspection and corrective-action tracking app covering 13,000 rooms: inspection cycles, findings, escalation rules, templated email with distribution lists, cron-based facility-data import; 140+ RSpec files from day one.
- Wrote the team's Okta OIDC single-sign-on Rails engine (PKCE, JWKS token verification, outage fallback, RP-initiated logout, per-app cutover preflight) and folded it into the shared auth engine used by every app.
- Migrated my applications from on-premises hosting to AWS, containerizing them with Docker and standing up GitHub Actions CI with test, lint, security and dependency checks.
- Modernized seven production apps from Ruby 3.4 / Rails 7 to Ruby 4 / Rails 8.1, then ran security and performance passes on each (N+1 fixes, static-analysis findings, Dart Sass migration, dead-code removal).

Software Engineer – Lead Developer, Massachusetts General Hospital **Dec 2023 – Jan 2026**

- Led the rewrite of the lab safety and compliance system (20,000 rooms, 4,000 PI sites, ~400 inspections a year) from an Ember front end into a single Rails app: inspection surveys and events, corrective actions and reminders, nightly facilities-data sync with retries, Pundit/Guardian permissions.
- Upgraded the evaluation, innovation-contest and suggestion apps from Rails 6 to Rails 7.1 / Ruby 3.3, clearing deprecations ahead of the Rails 8 move.
- Primary developer on the enterprise performance-evaluation system (~40,000 evaluations a year, 140,000+ logins over five months, 260,000 employee records, 30 institutions): employee search, a rewritten bulk employee-record importer and identity sync, optimized notification queries, and the PeopleSoft-to-Workday transition.
- Rewrote the enterprise suggestion platform (notification preferences, reminder schedules, role consolidation, status workflow, Microsoft Graph inbox integration) and migrated the suggestion-triage app to a new VM with a test-coverage push.

Software Engineer – Junior Developer, Massachusetts General Hospital **Dec 2021 – Dec 2023**

- Took over four production Rails apps, carried them through their first framework upgrades (Rails 5.1 / Ruby 2.6, then Rails 6) and shipped tracked features through code-reviewed pull requests from the first week: leaderboards, comments, status history, bulk suggestion mail.
- Owned evaluation workflow features: second-evaluator rules, reporting-structure violation notifications, per-institution templates, statistics exports and dashboards.
- Made the innovation-contest site (11,000 registered participants) configurable per contest: settings, validation, customizable submission and application pages, HoneyBadger error reporting.
- Built my first application solo from inception, a research grant portal handling funding rounds, application forms, review scoring, steering-committee access, Excel/PDF exports and external user accounts.

EDUCATION AND PRIOR EXPERIENCE

Actualize Coding Bootcamp – Certificate in Full Stack Web Development (2021) | **Keene State College** – B.A., Psychology (2013)

Ten years leading theatrical scenic and properties departments (2013–2021) at The Public Theater, The New School and Theatre for a New Audience, managing builds, budgets and crews on fixed deadlines.